

Brain alignment as a usable training signal: the internal extended manuscript

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Abstract

For a person and a language model reading the same text, a simple linear map predicts the person’s brain activity from the model’s middle-layer activations. For a decade this correspondence has been used in one direction only: to measure a model’s alignment with the brain after it is trained. We ask whether it can instead be used during training. We test this in the setting of model compression: we distill a language model into a smaller student with brain alignment in the objective, at a matched budget, and ask whether the student preserves or improves that alignment, a question that remains untested. An information-budget argument implies that such a signal is far too small to act as a second teacher; at most it is a weak prior. Such a prior should help only where the text signal is weakest (small models, scarce data, aggressive compression) and wash out wherever data is abundant. **[GAP: Results summary deferred (checkpoint 2): the two established findings (the powered reality result and the no-preservation distillation screen) and the downstream arc (per-individual null, averaging confound, quality law, predictability ceiling, no practical payoff) are summarised here only once R08–R14 are consolidated; close on the payoff line then.]**

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Notation and terms

Encoding model

A ridge regression from a model’s per-stimulus features to measured brain activity, scored out-of-sample. The standard brain-alignment instrument.

Voxel / ROI

A voxel is one 3D measurement location in an fMRI scan; an ROI is a predefined region (here, the five left-hemisphere language regions of the ROI screen).

ΔR^2 (unique R^2)

The variance the language-model features add *on top of* the nuisance set: $R^2([Z_{\text{nuis}}, \text{LM}]) - R^2([Z_{\text{nuis}}])$. The semipartial R^2 , not the partial. The only predictive score this work reports.

Trained–untrained gap

The verdict statistic: the unique R^2 of a language-trained model minus that of a same-architecture untrained network, under identical nuisance and splits. Isolates what *language training* adds over random initialisation.

Z_{nuis} (nuisance set)

The low-level confounds subtracted in both arms of the gap: utterance length, position, word and phoneme rate, word duration, log word-frequency, and a static word-embedding block (eng1000). Surprisal is deliberately excluded (it is itself LM-derived).

Noise ceiling

The maximum variance any model could explain, set by the reliability of the neural measurement itself; reported here as split-half reliability of a held-out repeated story.

Permuted twin

The per-kind shuffled-target null that tests brain-*specificity*. It belongs to the lever question (whether the signal can be *moved*), not to the reality question, and so does not appear in the Results section written here.

$\mathbb{E}[Y \mid S]$

The stimulus-predictable mean response: the strong conditioning variable behind the ceiling question, as opposed to the weak low-level proxy Z_{nuis} .

$\mathcal{L}_{\text{brain}}$

The brain-alignment training loss: the term added to a language objective that rewards representations predictive of recorded brain activity.

Knowledge distillation (KD)

Training a small student model to match a larger teacher’s output distribution rather than the raw text labels. Pure logit KD is the perplexity-only form (temperature-scaled KL on the teacher’s next-token logits, no hidden-state or brain term).

Cold-init / warm-init KD

A *cold-init* student is randomly initialised before KD, so any alignment it ends with was transmitted through the distillation channel; a *warm-init* student starts from a pretrained checkpoint and is therefore a fine-tuning-drift control, not a test of what KD carries.

Headroom (Δ)

The teacher-minus-student gap in unique R^2 , $\Delta = A_T - A_S$, with A_T the teacher’s alignment and A_S the student’s. The alignment a perplexity-only objective leaves below the teacher.

Floor-anchored retention (ρ')

The fraction of the teacher’s alignment a student keeps above the untrained floor, $\rho' = (A_S - A_0)/(A_T - A_0)$, where A_0 is the untrained score: $\rho' = 0$ is the floor (destroyed), $\rho' = 1$ is the teacher (preserved). Distinct from the partial correlation $\rho_{XY.Z}$ of the methods (the primed symbol is retention, never a correlation).

Data-processing inequality (DPI)

For a Markov chain $A \rightarrow B \rightarrow C$, no processing of B can increase information about A : $I(A; C) \leq I(A; B)$. Here it bounds a fixed-function compression to lose, never create, alignment relative to its teacher.

1 Introduction

A decade of work has established a simple, reliable instrument: for a person and a language model reading the same text, a linear map predicts the person’s recorded brain activity from the model’s middle-layer activations, and how well it predicts on held-out data is the model’s brain-alignment score [1]. It almost always runs in one direction: to *measure* alignment as a property of a model after training. We invert it. Can that alignment instead be *used* as a training signal, a term in the objective that rewards representations predictive of brain activity, and does a model trained that way gain anything a text-only model does not?

The answer turns first on an information budget. A language model already learns from on the order of 10^{12} tokens of dense, self-supervised supervision, while a language-fMRI dataset offers a few thousand noisy stimuli whose explainable variance is capped by a low noise ceiling. Adding a brain term to a language objective is, up to constants, a maximum-a-posteriori estimate in which the text enters as the likelihood and the brain term as a prior, not a second teacher. A prior worth, generously, 10^5 bits cannot install task knowledge against a likelihood worth 10^{12} ; what it can do is *select* among the many text-consistent solutions a large model admits, the way any weak-but-aimed regularizer does. Standard sample-complexity bounds then yield a falsifiable prediction: if a brain prior helps at all, it helps most where the text signal is weakest (small models, scarce data, aggressive compression), and washes out where data is abundant, the benefit shrinking as the task data grows. We make this argument formal in Section 3 and tie it to the information-theoretic bounds behind it.

Using fMRI to shape a language model is not new. Prior work does it for speech models [2], for text models [3], and causally: text models driven to be brain-*misaligned* at matched perplexity do substantially worse across a broad downstream suite, so alignment is causally

[DIAGRAM PLACEHOLDER: pipeline / method schematic]

Left-to-right setup, then the inversion. (a) A stimulus (a story or sentence) is read by a person and by a frozen language model; the model’s middle-layer activations X are extracted. (b) A ridge encoding map $\hat{W}$ predicts recorded brain activity Y (fMRI) from X ; alignment is the held-out unique variance $\Delta R^2(X; Y \mid Z_{\text{nuis}})$, the increment over a low-level nuisance floor, reported as the trained–untrained gap. (c) The inversion: the same alignment becomes a training term $\lambda \mathcal{L}_{\text{brain}}$ added to knowledge distillation, where a smaller student is trained from a teacher at a matched budget. Render in TikZ; keep three labelled stages (extract / measure / use).

Figure 1: The instrument and its inversion: how brain alignment is measured, and how it is turned into a training signal during distillation. **[GAP: Round-3: finalise caption once §3.6 (training term) is written.]**

relevant rather than incidental [4]. Every one of these interventions *grows* the model: it fine-tunes, adds adapters, or bolts on a brain-prediction head; none compresses one. The one study that compresses language models measures how quantization and pruning move an *observed* alignment score, never puts alignment in the training objective, and never tests knowledge distillation at all [5]. On two axes, what happens to the parameter budget (grow or shrink) and how alignment is treated (measured or optimized), the field fills three cells and leaves one empty: *shrink the model while alignment is in the objective*. The contribution of this thesis is to occupy that empty cell, alignment-guided distillation at a matched student budget, the regime the weak-prior argument singles out as the place to test whether a brain prior is recoverable at all rather than one where it is assumed to pay off.

The contribution depends entirely on the alignment score being a real signal rather than an artifact, because the field has repeatedly shown how easily it is not. Shuffled train-test splits leak through the temporal autocorrelation of fMRI, to the point that an *untrained* network can appear to “explain” brain data through sentence length and position alone [8]. We therefore fix the measurement standard up front: we never report a raw score, but the unique variance the model adds on top of a low-level nuisance set, on contiguous splits, with a same-architecture untrained network as the control [1, 9]. The program is then a sequence of dependent questions, each asked only if the one before it holds: whether the signal is real beyond confounds, whether plain distillation preserves it, whether an alignment-guided objective moves alignment beyond what perplexity already implies, whether any such gain holds per individual at matched perplexity, and whether it improves a downstream task.

Contributions. **[GAP: Round-3 itemised contributions, scoped through Q2 (no Q3–Q4 verdicts): (i) formulate brain-alignment-guided distillation and locate it as the empty cell; (ii) an information-budget theory predicting where a brain prior can help (the weak prior); (iii) a confound-clean measurement protocol (unique variance over a capacity-fair nuisance floor, contiguous splits, same-architecture untrained control, held-out noise ceiling, permuted-twin specificity); (iv) the powered reality result [E006]; (v) the no-preservation distillation screen [E003]; (vi) the weak, quality-**

[DIAGRAM PLACEHOLDER: 2×2 positioning of the field]

*A two-by-two grid. Rows = what an intervention does to the parameter budget (grow / shrink); columns = how it treats brain alignment (measured / optimised). Cells: (grow, measured) classical brain-score [1, 6, 7]; (grow, optimised) brain-tuning [2–4]; (shrink, measured) compression robustness [5]; (shrink, optimised) **this thesis**, starred, labelled “empty in the 2026 literature”. Render in TikZ from Table 1; visually emphasise the starred empty cell.*

Figure 2: The field on two axes; three cells are occupied and the fourth, where this work sits, is empty.

confounded lever [E004].]

We report results in the order the questions depend on one another. Two are established so far. First, the foundation everything else qualifies: a trained model’s middle layer carries unique, out-of-sample variance about real language-network activity that a same-architecture untrained network does not, and at voxel scale the effect is powered and unambiguous [E006]. Second, plain perplexity-only distillation does not preserve that alignment by default: a from-scratch student trained on the teacher’s outputs alone lands far below the teacher, though whether the drop is specific to the distillation objective or tracks the language-model quality the student loses is not resolved at this benchmark [E003]. *Real* does not imply *movable*, and *movable* does not imply *useful*: the first result establishes that the signal is real, and the second that plain compression does not keep it for free. **[GAP: Introduction summary of the remaining downstream findings (the per-individual null at matched perplexity, the cross-subject-averaging confound, the quality law, the predictability ceiling, no practical payoff) awaits the finding-reports R08–R14; do not state these verdicts here until their report layer exists.]**

2 Related work

The work on language models and the brain organizes on two axes, and on those axes three of the four cells are occupied while the fourth, where this work sits, is empty. The first axis is what an intervention does to the parameter budget: it either *grows* the model (fine-tunes, adds adapters, attaches a brain-prediction head) or *shrinks* it (quantizes, prunes, or distills into a smaller student). The second is how it treats brain alignment: either as a quantity *measured* after the fact, or as one *optimized* inside the training objective. Three of the four cells are occupied; the fourth is empty, and it is the thesis (Table 1).

Table 1: The field on two axes: what an intervention does to the parameter budget, and how it treats brain alignment. Three cells are occupied; the fourth, shrinking a model while alignment is in the objective, is empty in the 2026 literature and is where this work sits.

	Alignment measured	Alignment optimized
Model grown	classical brain-score [1, 6, 7]	brain-tuning [2–4]
Model shrunk	compression robustness [5]	this thesis (empty in the literature)

2.1 Grown models, alignment measured: the classical brain-score literature

The largest body of work fits encoding models to frozen, pretrained networks and reads off how alignment varies with architecture, scale, and training. Alignment is carried by structured middle-layer representations, with syntactic structure driving the layerwise trend and semantics more region-specific [1]. It rises with model scale and saturates in the few-billion-parameter range; at matched parameter count, scale rather than instruction-tuning moves it, which is why a base model is the appropriate alignment teacher [6]. Past that range the relationship can decouple and even reverse, as the largest models outgrow the human language network and track formal linguistic competence more than behavior [7]. This cell establishes that the signal exists and is non-trivial, but it never intervenes through alignment, so it cannot speak to whether alignment is usable.

2.2 Grown models, alignment optimized: brain-tuning, and why the bare inversion is not novel

Several studies put alignment in the objective of a fine-tuned or grown model, across modalities. For speech, a voxelwise loss fine-tunes compact encoders and improves both alignment and several downstream tasks, with a multi-participant variant reporting further gains at higher fMRI-data efficiency [2, 10]. For text, fMRI-augmented training of GPT-2 and LLaMA-2 with a cosine-similarity brain loss beats text-only baselines, though the reported gains are mixed across models and the splits are random rather than contiguous with no shuffled-brain control, so they would not pass the anti-confound bar adopted here [3]. Causally, driving text models *away* from alignment at matched perplexity damages performance across a 200-plus-task suite, and the same study’s positive arm shows that driving toward alignment helps, evidence that the recoverable capability is real [4]. The bare inverted idea is therefore not novel; what no study in this cell does is compress. Every one of them grows or holds the parameter budget, which leaves the compression question entirely open.

2.3 Shrunk models, alignment measured: the compression study that does not close the gap

One study compresses language models and tracks alignment, and it is the closest prior work to this thesis, so its scope needs a precise reading. It finds that alignment survives only particular post-training quantizers on models of at least a few billion parameters; other quantizers, smaller models, and pruning all degrade it, and it never tests knowledge distillation [5]. Critically, this is *shrink* crossed with *measure*, not with *optimize*: it observes alignment under a fixed-function perturbation of the weights and never makes it a training target. Distillation is a different and lossier operation: it re-fits a smaller function from scratch against the teacher, the regime most likely to discard the middle-layer structure where alignment lives, so a result about quantization does not transfer to it. The data-processing inequality frames why the question is quantitative: a student can only lose alignment relative to its teacher, never create it, and in the narrow regime where preservation holds (particular quantizers on models of at least a few billion parameters)

that bound is nearly tight; the open question is whether distillation, a lossier transform, sits well below tightness by default and whether a brain objective pulls the student back toward the bound.

2.4 The empty cell, and the adjacent precedents that motivate it weakly

The remaining cell, shrink the model while alignment is optimized at a matched student budget, is unoccupied, and it is where this work sits. Two precedents from vision and audio are sometimes read as support for a brain regularizer in the low-data regime, but each weakens under scrutiny. A neural-data regularizer on a vision network improves accuracy and adversarial robustness, but a shuffled-label control reproduces the accuracy gain, so only the robustness benefit there requires real neural structure [11]. fMRI fine-tuning of a compact audio network improves a majority of downstream tasks, but the apparent low-data advantage carries no interaction test and no matched non-brain control, so it is suggestive rather than isolated [12]. These precedents are methodological cautions more than positive evidence: isolating the brain-specific increment requires the matched non-brain control both studies omitted, and neither supplies a result that transfers cleanly to the distillation regime.

2.5 The measurement-validity bar, and the fMRI-free surrogate question

A parallel line argues that brain-alignment scores are easy to inflate and easy to over-read, and it sets the standard this work measures against. Shuffled splits leak through temporal autocorrelation, low-level and static features carry much of the apparent signal, and audited frozen-encoding scores can fail basic predictive and reliability gates [8, 13]. A related caution is that aggregate scores hide mechanism: the same headline number can be genuine semantics in one model and shallow phonology in another [14]. A more ambitious route would replace the live fMRI loss with a differentiable surrogate, but the literature does not yet agree on which way to push one: local intrinsic dimension correlates with alignment in opposite directions across language and vision studies, and a similarity-based surrogate is confounded by width and depth, with only neighborhood-overlap surviving null calibration [15–17]. A surrogate objective is therefore held as future work, since its very direction must first be validated against real fMRI.

3 Methods

3.1 The encoding model and the alignment score

We measure alignment with an encoding model: a ridge regression from a frozen model’s per-stimulus middle-layer activations X to recorded brain activity Y , fit on training stimuli and scored on held-out ones. For activations $X \in \mathbb{R}^{n \times d}$ over n stimuli and a v -voxel response Y , the fitted map is

$$\hat{W} = \arg \min_W \|Y - XW\|_F^2 + \lambda \|W\|_F^2, \quad (1)$$

and the per-voxel score is the held-out correlation between predicted and measured response. We do not report a raw score of this kind on its own, because a model can predict the brain for reasons that have nothing to do with language understanding: utterance length, word position, generic word identity. We therefore report the unique variance the model adds *on top of* a low-level nuisance set Z_{nuis} , the difference of two cross-validated fits,

$$\Delta R^2 = R^2([Z_{\text{nuis}}, \text{LM}]) - R^2([Z_{\text{nuis}}]), \quad (2)$$

a positive value of which says the contextual representation explains brain activity that length, rate, and static word identity cannot [E002].

3.2 What that quantity is, formally

The score is a lower bound on the information the model’s representation carries about brain activity beyond the low-level confounds. The unique-variance score is, formally, the linear-Gaussian estimator of a conditional mutual information. For a triple (X, Y, Z) , the conditional mutual information $I(X; Y | Z) = H(Y | Z) - H(Y | X, Z)$ is the dependence between X and Y that survives once the context Z is known [18]. It is the right object because plain mutual information conflates a genuine dependence with a spurious one in which X and Y are merely co-driven by a common source: when that holds, $X \perp Y | Z$ and the conditional term is zero even though the unconditional one is positive, so conditioning is what tells the two apart. The chain rule splits the question into a baseline and an increment, $I((X, Y); Z) = I(X; Z) + I(Y; Z | X)$, so the value of a learned representation Y over a cheap baseline X is the increment $I(Y; Z | X)$. Identifying the nuisance with X , the model’s middle layer with Y , and brain activity with Z , the reality claim is that $I(\text{LM}; B | Z_{\text{nuis}}) > 0$, which the unique-variance score estimates with ridge. For jointly Gaussian variables the conditional mutual information is a monotone function of the squared partial correlation, $I(X; Y | Z) = -\frac{1}{2} \ln(1 - \rho_{XY \cdot Z}^2)$ [18, Ch. 8]. The reported score is the *semipartial* R^2 , the raw increment, which differs from the partial ρ^2 by a factor $1/(1 - R^2([Z]))$; both vanish together, so a positive semipartial score is equivalent to a positive conditional mutual information, which is the biconditional the argument needs. Because ridge is linear, the score captures the linearly decodable part of that dependence: a population-level lower bound on the true information, and the appropriate target for a thesis about the linear map.

The choice of conditioning variable scopes the whole program. Conditioning on the low-level nuisance Z_{nuis} establishes that the signal is real beyond low-level confound, not beyond the stimulus as a common cause; the stronger conditioning on the full stimulus-predictable response $\mathbb{E}[Y | S]$ is a separate, later question. The same chain rule with a stronger Z is what the downstream ceiling analysis evaluates.

3.3 Why brain data can only be a weak prior

The regime where a brain signal could matter follows from an information budget, and that budget predicts where the experiments should and should not find an effect. Adding a brain term to a language objective is, up to constants, a maximum-a-posteriori estimate in which the text enters as the likelihood and the brain term as the log-prior. The likelihood’s pull scales with the supervision it carries, on the order of 10^{12} to 10^{13} bits across a training corpus, while the brain side offers 10^3 to 10^4 noisy stimuli whose per-voxel information is capped by a noise-ceiling channel capacity $I \leq \frac{1}{2} \log_2(1 + \text{SNR})$ with $\text{SNR} = r_{\text{max}}^2 / (1 - r_{\text{max}}^2)$, putting the usable brain budget, as an order-of-magnitude estimate, around 10^4 to 10^6 bits [18]. The data-processing inequality shrinks even that: the brain response is a noisy function of the same stimulus the model already trains on, so for the chain $S \rightarrow R(S) \rightarrow Y$ it carries no more about the optimal parameters than the stimulus does, $I(\theta^*; Y) \leq I(\theta^*; S)$ [18]. A prior this weak cannot teach new task structure against a likelihood this strong; what it can do is *select* among the high-dimensional manifold of near-equivalent text-optima that a large model admits, which is the precise sense in which a brain term is a regularizer rather than a teacher. A generalization bound makes the consequence quantitative: a prior concentrated on the right hypotheses tightens a sample-complexity bound by an amount that scales as the square root of the extra information it injects over the task data, and so vanishes as that data grows [19]. A weak-but-aimed brain prior should therefore help most in low-information regimes (small, under-trained, or aggressively compressed models) and wash out where the text signal already dominates, which is the regime logic behind targeting distillation.

**[DIAGRAM PLACEHOLDER: the unique-variance score and the
trained–untrained gap]**

A small construction diagram. Top: for one network, two nested cross-validated ridge fits, $R^2([Z_{\text{nuis}}])$ and $R^2([Z_{\text{nuis}}, \text{LM}])$; their difference is the unique-variance score ΔR^2 (the increment the representation adds over the nuisance floor). Bottom: the same construction run twice, once for the trained network and once for the same-architecture untrained one; the reported statistic is the gap, trained ΔR^2 minus untrained ΔR^2 , so the shared nuisance term cancels. Render in TikZ as two stacked bar-pairs with the increment and the gap annotated.

Figure 3: What the reported number is: the unique variance a network adds over the nuisance floor, and the gap between the trained network and its untrained twin.

3.4 The verdict statistic: the trained–untrained gap

We report not a trained model’s absolute unique-variance score but the *gap* between it and a same-architecture untrained network fit under identical nuisance and splits. The reason is that on naturalistic data an untrained network can score slightly positive through a shared-cause path: the stimulus drives both the response and the untrained features, leaving a residual dependence when the nuisance only partially captures the stimulus, so the absolute score is not a clean measure of what training adds. Both arms pass through the byte-identical nuisance subtraction, so the baseline term cancels in the difference and the gap directly isolates what *language training* contributes over random initialisation, controlling for architecture, tokenisation, and residual temporal structure [E006]. We report this gap throughout.

3.5 Three controls, fixed as the standard

Three controls make that gap trustworthy, and we apply them uniformly. *Contiguous splits*: train and test are divided in blocks (block cross-validation on the ROI screen, whole-story-held-out cross-validation on the voxelwise run), because shuffling time-adjacent fMRI samples leaks near-duplicate timepoints across the split and inflates the score [E002][E006]. *A capacity-fair nuisance floor*: the wide contextual block and the wide static-embedding block are each reduced by PCA to the same rank, fit on the training fold only, so that when the model wins it wins on content rather than on carrying more dimensions; the few low-level scalars stay raw [E006]. *A held-out noise ceiling*: on the voxelwise run only reliable voxels are kept, and they are selected on a separate story repeated ten times and held entirely out of the cross-validation pool, scored by split-half reliability, so the estimate is not inflated by voxels that happen to correlate with the model by chance [E006]. Surprisal is deliberately left out of the nuisance set, since it is itself an LM-derived quantity and subtracting it would over-subtract the construct under test; log word-frequency, a static lexical scalar, is distinct and is subtracted [E006]. The permuted-target null (brain responses shuffled across stimuli, which breaks the stimulus-response pairing while preserving temporal structure) tests brain-*specificity*, not reality, and enters only where the question is whether the signal can be moved.

3.6 Data and models

Two real datasets span the design. The ROI screen uses 1000 isolated sentences and the five left-hemisphere language regions of Tuckute et al. [20], a coarse but high-ceiling, train-participant-averaged benchmark whose isolated-sentence design sidesteps the temporal-autocorrelation leakage of naturalistic time series; its reported noise ceiling is $r \approx 0.35$ [E002]. The powered measurement uses subject UTS03 of the deep-sampled naturalistic story-listening dataset of LeBel et al. [21], with 20 stories, story-level folds, and 11,442 voxels retained at split-half reliability above 0.5 on a held-out repeated story [E006]. The models are GPT-2, GPT-2-medium, and Qwen2.5-0.5B, each read at its alignment-peak middle layer, with a same-architecture untrained network as the control in every comparison [E002][E006].

3.7 Distillation setup and the brain training term

[GAP: Round-3: write the distillation + $\mathcal{L}_{\text{brain}}$ training setup. The mechanism it describes is the bottom (“use”) stage of Figure 1; cross-reference rather than redraw.]

4 Results

4.1 The alignment signal is real, beyond confounds

A trained model’s middle layer carries unique, out-of-sample variance about real language-network activity that a same-architecture untrained network does not, and the effect survives the full low-level nuisance regression under contiguous splits. The ROI screen passes on every model tested, and the powered voxelwise run confirms it where the screen could only hint.

The ROI screen, on the five left-hemisphere language regions of Tuckute et al. [20] with 1000 isolated sentences over five contiguous folds, shows a positive trained-minus-untrained gap for all three models, with the untrained control negative at every layer and seed (Table 2). The signal is positive across a broad plateau of middle layers rather than at a single peak, and is absent from random features [E002].

Table 2: ROI screen on Tuckute 2024 (five LH language ROIs, five contiguous folds). Unique R^2 is the semipartial increment over the nuisance set; the gap is trained minus untrained. NC-norm is the gap as a fraction of the $r \approx 0.353$ noise ceiling.

Model	Best layer	Trained ΔR^2 (mean $\pm$ SD)	Untrained ΔR^2	Gap	NC-norm	Source
GPT-2 (12L)	L7	$+0.020 \pm 0.008$	-0.011	$+0.030$	0.056	[E002]
GPT-2-medium (24L)	L14	$+0.019 \pm 0.005$	-0.014	$+0.033$	0.053	[E002]
Qwen2.5-0.5B (24L)	L12	$+0.036 \pm 0.010$	-0.014	$+0.050$	0.102	[E002]

The powered voxelwise run, on subject UTS03 of LeBel et al. [21] over 20 naturalistic stories and 11,442 voxels selected at split-half reliability above 0.5 on a held-out repeated story, gives a trained-minus-untrained gap of $+0.0207$ [95% CI $+0.0205$, $+0.0209$] for GPT-2 and $+0.0277$ [95% CI $+0.0274$, $+0.0280$] for Qwen2.5-0.5B, with 95% and 99% of reliable voxels positive respectively and the untrained unique R^2 at -0.017 in both (Table 3) [E006]. The confidence intervals exclude zero, and the gap is carried by language training rather than by architecture or the temporal structure the nuisance floor absorbs [E006].

Table 3: Powered voxelwise run on LeBel UTS03 (20 stories, story-level folds). The gap is the trained-minus-untrained unique R^2 ; CIs are bootstrap.

Model	Layer	Trained – untrained gap [95% CI]	% reliable voxels gap > 0	Source
GPT-2	L7	+0.0207 [95% CI +0.0205, +0.0209]	95%	[E006]
Qwen2.5-0.5B	L12	+0.0277 [95% CI +0.0274, +0.0280]	99%	[E006]

Three controls rule out the obvious artifacts, and the screen and the powered run are complementary on them. The ROI screen uses isolated sentences, which removes the temporal-autocorrelation inflation that contaminates naturalistic time series, so its positive is harder to obtain rather than easier [E002]. The voxelwise run reintroduces naturalistic stimuli but adds whole-story-held-out cross-validation, so test timepoints are never adjacent to training timepoints, and voxel selection on a separate held-out repeated story, so the score cannot be inflated by voxels that correlate with the model by chance [E006]. The nuisance set subtracted in both arms is the low-level set most likely to manufacture a false positive: word and phoneme rate, word duration and length, log word-frequency, and a static word-embedding block [E006]. Surprisal and imageability are deliberately left unsubtracted, so a residual contextual nuisance is not ruled out by the floor alone; what rules it out is the same-architecture untrained network, which sees the identical set and scores negative at -0.017 , so architecture, rate, and residual temporal structure cannot manufacture the gap [E006].

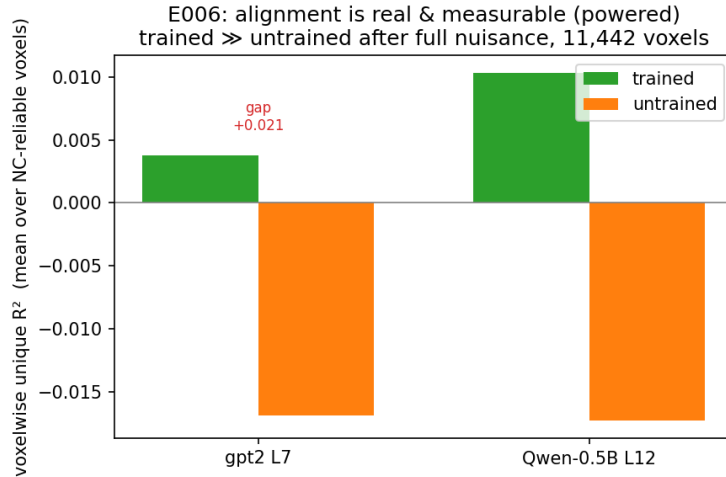


Figure 4: The powered reality result: the trained-minus-untrained unique- R^2 gap is positive across the overwhelming majority of reliable voxels on LeBel UTS03 [E006].

The signal is real beyond confounds on real neural data. The three-model ROI screen carries the breadth, and the deeply-sampled single subject UTS03 carries the statistical power at voxel scale. Cross-subject generality of the powered effect is a question this single-subject design leaves open, not one it settles. Qwen aligns more strongly than either GPT-2 variant, while the two GPT-2 variants are nearly tied despite a threefold parameter difference, so this single cross-family contrast does not by itself establish a quality ordering; that question is taken up later [E002]. This is the foundational positive in the program: necessary but not sufficient for everything after it. *Real* does not imply *movable*, and *movable* does not imply *useful*: this result establishes only that the signal is real.

4.2 Plain distillation does not preserve alignment by default

Knowledge distillation (KD) trains a smaller student to match a larger teacher’s output distribution rather than to learn directly from text. A student trained that way does not keep the teacher’s brain alignment for free: ordered by how much of each representation was rebuilt through the distillation channel rather than inherited from ordinary pretraining, unique R^2 falls monotonically, and a from-scratch logit-KD student lands far below its teacher, a gap of +0.0181 ($p < 0.001$) at which it retains only 0.37 [0.14, 0.58] of the teacher’s alignment above an untrained floor [E003]. This rules out the outcome that would have made the compression question a non-problem, preservation by default, and sizes the alignment a perplexity-only objective leaves below the teacher. What it does not establish is the stronger claim that the distillation objective sheds alignment beyond the language-model quality it costs: across the trained and distilled models alignment tracks log-perplexity at Pearson -0.88 , so quality alone could account for the whole gradient [E003][L011].

The quantity is the headroom $\Delta = A_T - A_S$ between the teacher’s alignment A_T and the student’s A_S , both measured on the apparatus of the reality result: unique R^2 on the Tuckute ROI screen, contiguous folds, the capacity-fair nuisance floor. The gap is an empirical question rather than a corollary, and the reason fixes the design. For a deterministic post-processing of a fixed teacher, the quantization and pruning of Oota et al. [5], the student’s features depend on the stimulus only through the teacher’s, so the data-processing inequality bounds $I(Z_{\text{student}}; B) \leq I(Y_{\text{teacher}}; B)$ and a fixed-function compression can only lose alignment, never create it; there the scientific content is how much is lost, not whether. Distillation is not that regime: a from-scratch student fit by gradient descent on the teacher’s logits has a side-channel to the stimulus through the raw next-token targets that bypasses the teacher’s features, the data-processing premise fails, and nothing forces the student back to the teacher’s alignment [E003].

The verdict therefore rests on a cold-init student. A warm-started student inherits full teacher-grade alignment from its own pretraining, which the reality result already measures, so a high retention for it would show only that a few KD epochs did not erase the alignment the student already had, not that the KD channel carried it [E003]. The verdict-bearing arm is a randomly initialised GPT-2 trained on the teacher’s logits alone, the one arm whose surviving alignment was transmitted through distillation; we keep a warm-init KD arm and a no-teacher plain-LM-finetune arm as labelled fine-tuning-drift and corpus-drift controls, and we score every arm at a fixed layer within the 12-layer GPT-2 architecture so no cross-architecture layer choice confounds the comparison [E003].

Table 4: Knowledge-distillation arms on the Tuckute ROI screen at the fixed verdict layer (NC-normalised unique R^2 , noise ceiling 0.353) [E003]. Floor-anchored retention $\rho' = (A_S - A_0)/(A_T - A_0)$ is 0 at the untrained floor and 1 at the teacher; Δ is the gap versus the teacher; $p(\text{no-drop})$ is the bootstrap probability of no drop from the teacher.

Arm	Init / objective	ΔR^2	ρ' [boot. CI]	Δ vs teacher	$p(\text{no-drop})$	ppl	Source
teacher gpt2-medium	pretrained	+0.0188	n/a	n/a	n/a	74	[E003]
gpt2	pretrained (conventional)	+0.0195	1.02 [0.77, 1.27]	≈ 0	0.57	105	[E003]
lmft_warm	warm + plain-LM finetune	+0.0174	0.95 [0.81, 1.13]	+0.0016	0.265	44	[E003]
kd_warm	warm + pure logit KD	+0.0144	0.84 [0.67, 1.04]	+0.0046	0.061	95	[E003]
distilgpt2	real distillation (+hidden-cosine)	+0.0076	0.60 [0.42, 0.84]	+0.0119	0.001	169	[E003]
kd_cold	random + pure logit KD	+0.0005	0.37 [0.14, 0.58]	+0.0181	0.000	477	[E003]
untrained	random (floor)	-0.0102	0	n/a	n/a	~ 50000	[E003]

The durable finding is the monotone gradient (Table 4). Retention falls cleanly along the rebuild ordering: conventional GPT-2 at the teacher’s level ($\rho' \approx 1.0$), warm-init KD at 0.84, the published distilgpt2 at 0.60, the from-scratch cold-init student at 0.37, and the untrained floor at 0 [E003]. The reference arms validate the measurement: off-the-shelf GPT-2 sits at the

Missing figure: ../figures/fig_kd_retention.png

Figure 5: [PLACEHOLDER: generate from [E003]] Two panels. (A) Floor-anchored retention ρ' along the rebuild ordering (teacher, conventional GPT-2, warm-init KD, distilgpt2, cold-init KD, untrained floor) with bootstrap CIs, showing the monotone fall. (B) Unique R^2 versus log-perplexity across the same arms, with the Pearson -0.88 fit, showing that alignment co-varies with model quality. **[GAP: Round-3: render the two-panel figure (scripts/figures) and finalise the caption.]**

teacher’s level and the untrained control at the floor, and two runs scoring the same references on different GPUs reproduced them to about 0.0003 [E003]. The ordering holds at every PCA rank tested ($n_{\text{pca}} \in \{25, 50, 100\}$), but the absolute shed fractions are rank-sensitive and the two confounded arms flip sign across ranks, so the ordering is the finding and the magnitudes are rank-dependent [E003].

The gradient does not isolate the distillation objective from the language-model quality the student loses, because alignment co-varies with quality across the same arms. Alignment is tightly predicted by log-perplexity across these arms (Pearson -0.88). On that fit, warm-init KD lies essentially on the regression line, and conventional GPT-2 is the lone point above it, the only arm more aligned than its perplexity predicts. The two contrasts that would separate an objective-specific loss from “alignment tracks quality” are both only marginal: warm-init KD is less aligned than GPT-2 despite lower perplexity, but the bootstrap $P(\text{gpt2} \leq \text{kd_warm}) = 0.092$, and at matched budget the no-teacher finetune beats warm-init KD on both perplexity and alignment, but $P(\text{lmft} \leq \text{kd_warm}) = 0.085$; each leans on a single unreplicated point and, under a paired-fold test, on one fold carrying about 44% of the signal [E003][L011]. The cold-init arm compounds the difficulty with under-training, its held-out perplexity (477) sitting far above the converged arms’ (44 to 95), so part of its near-floor alignment is simply a worse language model rather than distillation shedding alignment, the convergence caveat the design predeclared [E003]. The match across arms was the compute budget, not the optimizer-step count: the from-scratch arm ran about twice the steps of the warm arms, which deepens this caveat rather than flattering it [E003].

Plain perplexity-only distillation does not preserve brain alignment by default, and the headroom below the teacher is genuine but quality-confounded. Whether an alignment-guided objective can recover that headroom beyond what perplexity already implies is the matched-perplexity question the rest of the program takes up, not one this screen settles. The benchmark here is a cheap ROI screen ($\text{NC} \approx 0.35$) at matched compute budget, under-powered to resolve a ~ 0.005 objective-specific gap and not run at matched perplexity, which is why that question moves to the powered voxelwise substrate [E003][L011].

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Figure 6: [PLACEHOLDER: generate from [E004]] The brain-specific lever on the strong aligner: held-out unique- R^2 change from optimising $\mathcal{L}_{\text{brain}}$ against its permuted twin control, shown at the honest fold-level inference unit (folds, not pooled cells), with the interval spanning zero and the single dominant fold marked. **[GAP: Round-3: render from the E004 record and finalise the caption.]**

4.3 An alignment-guided objective moves alignment, weakly and confounded by quality

[GAP: Results Q2 (source E004), to write in Round 3: optimising $\mathcal{L}_{\text{brain}}$ on the strong aligner moves held-out alignment only weakly; the effect is confounded by perplexity; at the honest fold-level inference unit the interval includes zero, so no lever is demonstrated (undemonstrated, not proven zero).]

4.4 No per-individual gain at matched perplexity, and the averaging confound

[GAP: Results Q3 awaits finding-reports R09–R13 (sources E005, E008, E011, E013, E017, E019, E020): the per-individual null, the cross-subject-averaging confound, the method-general failure, the quality law, the stimulus-predictability ceiling, and the external reproduction. The parked draft reports/_pending-Q3 reclaims R09.]

4.5 No practical payoff

[GAP: Results Q4/A3 awaits finding-report R14 (source E009): the bounded OOD-perplexity null and its null-by-construction reading.]

5 Discussion

[GAP: Discussion awaits the consolidated Results arc (R07–R14). The intended spine: the positive contribution is a measurement-validity finding plus a confound-clean protocol, not a bare null; the cross-subject-averaging confound manufactures apparent brain-specificity, and the matched-perplexity, per-kind permuted-twin, per-subject protocol is the control the alignment-training literature lacks. Write only once the Results sections it interprets exist.]

6 Limitations

[GAP: Limitations awaits the consolidated Results arc. The intended content, each tied to its result: the absolute unique variance is a few percent of the noise ceiling (a scale fact, not a weakness); the reality claim is "beyond the nuisance actually subtracted," with surprisal held out by design; the per-individual null is measured at one power level on one cohort; the stimulus-predictability ceiling is instrument-limited rather than closed. Write once the results exist.]

7 Conclusion

[GAP: Conclusion awaits the consolidated Results arc. It should restate the established positive (the alignment signal is real and powered beyond confounds) and the program's net finding once R07–R14 are consolidated, without reaching past the report layer.]

Checkpoint log

2026-06-16: Checkpoint 1, the manuscript is opened

This is the first checkpoint: the extended manuscript is created and the layers that already trace to a written finding-report are drafted. The science behind it was produced over the preceding working sessions and recorded in the experiment files; this checkpoint consolidates the part of that record that the report layer has so far narrated.

What is folded in at this checkpoint. The measurement apparatus and the one established positive result: a trained language model's middle layer carries unique variance about real language-network activity beyond a low-level nuisance regression, and a same-architecture untrained network does not [E006]. This is consolidated from the finding-report on the reality of the signal (R06), which itself rests on the ROI screen [E002] and the powered voxelwise run [E006]. The front matter (Introduction, Related work) is consolidated from the background reports: the first-principles case for the brain as a weak prior (R03), the gap analysis that locates the contribution in the empty *shrink* $\times$ *optimize* cell (R04), and the synthesis of the measurement literature (R01).

What is deliberately left open. The Results sections beyond the reality of the signal (whether plain distillation preserves alignment, whether the loss is a usable lever, the per-individual null, the quality law, the predictability ceiling, and the practical-payoff question) are marked with \gap because their finding-reports (R07–R14) are not yet written. Knowledge flows down only: a number is narrated in a report before it is consolidated here, so these sections wait on the report layer rather than reaching past it into the raw experiment records. The abstract is provisional for the same reason: it can state the question and the reality result, but not yet summarise an arc whose later findings have no narration layer.

How this checkpoint was built. The draft was written through the scientific-writing skill on the full loop, with the review pass run continuously: adversarial critique agents argued against the reasoning and the prose quality section by section, and the objections that held were addressed before the section was kept.

2026-06-22: Checkpoint 2, the distillation-preservation result and a register repair

This checkpoint does two things, neither of which changes a recorded number: it folds in the next finding-report, and it repairs the prose register the first checkpoint shipped.

What is folded in. The Q1 result, that plain perplexity-only knowledge distillation does not preserve a teacher’s brain alignment by default, is consolidated from its finding-report (R07, source [E003]) into the second Results subsection. A from-scratch logit-KD student lands a gap of +0.0181 below its teacher ($p < 0.001$, retaining 0.37 [0.14, 0.58] of the teacher’s alignment above the untrained floor), and unique R^2 falls along a monotone gradient ordered by how far each representation was rebuilt away from conventional pretraining [E003]. The stronger claim, that the distillation objective sheds alignment beyond the language-model quality it costs, is left unsupported in the prose, because alignment co-varies with log-perplexity at Pearson -0.88 and the two objective-specific dissociations are only marginal [E003][L011]. The abstract and Introduction are updated to state this second result with that hedge, and their downstream-summary \gap markers are narrowed from R07–R14 to R08–R14. The glossary gains the terms the new section needs (knowledge distillation, cold-init/warm-init, headroom Δ , floor-anchored retention ρ' , the data-processing inequality); the retention symbol ρ' is disambiguated from the partial correlation $\rho_{XY.Z}$ of the methods, a notation collision the consolidation guard flagged.

The register repair. A supervisor reading the v0.1 body flagged a storytelling register that the writing-style rules forbid (L054, D046): abstractions handed verbs of agency, the rhetorical move narrated instead of made, the repo’s internal scaffolding metaphors leaking into reader-facing prose, and a reflex passive density above the threshold in the Introduction and Related work. This checkpoint rewrites those passages in the Introduction, abstract, Related work, and Results to assert the content directly, with a subject that acts and a verb that states, and varies the passive where the agent is the real topic. No claim, number, or hedge changed under the rewrite; only the voice did.

What is still open. The Discussion, Limitations, and Conclusion remain \gap because the results they interpret (the per-individual null, the averaging confound, the quality law, the ceiling, the practical-payoff question) await their finding-reports R08–R14. R07’s own caveats are carried inside the Results prose, as the report convention requires, rather than opened as a partial Limitations section ahead of the consolidated arc.

How this checkpoint was built. Written as its own logged /write session on the full loop (the ship gate D046 forbids folding a manuscript revision into another session), with the deterministic checks run on each edit and the non-skippable review pass, the prose-register-auditor included, run before handoff.

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